

the value-addition provided by the EMS partner and evaluate the EMS in terms of its ability to handle surprises and crises.

Where does the country lack that has prevented the EMS sector from growing and competing with global competitors?

Investments in developing capabilities to manufacture in India are a big roadblock that holds us back. Peripherals like loaders, conveyors, unloaders, bare PCB cleaners, storage accessories, and ESD-related materials are the basic necessities in every manufacturing setup. Thus, more investment is required to acquire and manufacture peripheral equipment related to SMT and TH (through-hole) lines.

We are completely dependent on the overseas market for these equipment. If we can acquire investment for these lines, we can truly be self-reliant and can also create a great opportunity for the export market. Similarly, India is dependent on the overseas markets for copper-laminated boards. These laminates, if produced on a larger scale, will reduce costs and also our dependency on imports.

Next, we have accessories like USB connectors, adaptors, pins, SIM holders, miniature switches, and related products imported in huge numbers. Needless to say, chip components like resistors and capacitors should also be made in India.

These components can be manufactured in India without any new technology. These can be handled in India, provided we have the hand-holding from the government till the industries become self-sustainable. We can expand our horizon to the export market once we stabilise in these areas.

Are there any global trends that Indian EMS is yet to catch on but would greatly benefit from?

Organic electronics can benefit

with potential low cost and high mechanical flexibility, even though they have low thermal stability. Also, different electronic components can be printed on top of each other, increasing reliability. They operate on low energy and pave way for creative products made from non-toxic materials at low cost. India is slowly catching up on this trend though the industry is growing comparatively faster worldwide.

What is your outlook for the future in terms of demand for EMS?

The trend is for IoT based products, electric vehicles and related accessories, and communication-related products. We expect this trend to continue for a long time.

Many design houses are turning into ODMs and some EMS firms set up a design centre too; what are your thoughts on this trend?

This is a good trend that is followed worldwide and provides a good scope for innovation and testing. Customers always prefer to have 'one number to ring' for all their needs, so this combination of design houses and EMS is a welcome change. This broadens the scope for EMS and helps in mitigating risks of dependency on customer's success with their designs. We too have partnered with design houses and are looking for opportunities on these lines.

Do you have any strategy to cater to the startups, who often find it difficult to connect with EMS firms to manufacture their prototypes in small quantities?

We are always forthcoming in supporting startups. Our teams visit incubation centres and startup parks to understand requirements of the new firms. Some of our existing customers are with us from their startup stage and we have grown together.

Moreover, we do have a prototype assembly line which caters to small

numbers. Engaging startups is a separate line of focus and is a separate business vertical.

With the rising cost of the US dollar and freight, do you see the possibility of an Indian EMS firm being able to compete with Chinese and other APAC counterparts any time soon?

We need to ask ourselves the question of whether we have what it takes to challenge China.

During Covid it became apparent that the monopoly in the supply-chain poses major risk. With ease of business widening in India, we are sure to grow as an influencer in electronics manufacturing in the next five years.

To make supply-chains more resilient and diversified, global customers have to look at India as an option.

This growth and exports will strengthen our currency against the US dollar. Also, it can enable better transporting hubs and solve the freight gaps too. Fast tracking freight is very much on our government's radar. We may soon witness cost-effective solutions. With the PLI schemes, India is competing with China in capturing the global market.

India has already captured the mobile phone and TV manufacturing to a great extent. The laptop and computer manufacturers should also consider India as the manufacturing destination. We are fast growing in the IoT based modules and are going to surpass China in this too.

Last but not the least, what role can media like Electronics For You play to enable India's EMS sector and overall electronics manufacturing sector to grow fast?

EFY can support the tough and strained sourcing department of EMS by providing integrated and verified market links. They can also act as an escrow account for supporting EMS players. Skill development and trained manpower availability for the industry can be supported by EFY. **EFY**